

Research and Applications

Design of patient-facing immunization visualizations affects task performance: an experimental comparison of 4 electronic visualizations

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Abstract

Objective: This study experimentally evaluated how well lay individuals could interpret and use 4 types of electronic health record (EHR) patient-facing immunization visualizations.

Materials and Methods: Participants ($n = 69$) completed the study using a secure online survey platform. Participants viewed the same immunization information in 1 of 4 EHR-based immunization visualizations: 2 different patient portals (Epic MyChart and eClinicWorks), a downloadable EHR record, and a clinic-generated electronic letter (eLetter). Participants completed a common task, created a standard vaccine schedule form, and answered questions about their perceived workload, subjective numeracy and health literacy, demographic variables, and familiarity with the task.

Results: The design of the immunization visualization significantly affected both task performance measures (time taken to complete the task and number of correct dates). In particular, those using Epic MyChart took significantly longer to complete the task than those using eLetter or eClinicWorks. Those using Epic MyChart entered fewer correct dates than those using the eLetter or eClinicWorks. There were no systematic statistically significant differences in task performance measures based on the numeracy, health literacy, demographic, and experience-related questions we asked.

Discussion: The 4 immunization visualizations had unique design elements that likely contributed to these performance differences.

Conclusion: Based on our findings, we provide practical guidance for the design of immunization visualizations, and future studies. Future research should focus on understanding the contexts of use and design elements that make tables an effective type of health data visualization.

Key words: data visualization; vaccination; child, patient portals; health records; personal.

Background and significance

Patients, because of legislative action, are now required to have electronic access to their personal health data in their medical record.¹ Patients can access these data in numerous ways, including via electronic health record (EHR) patient portals, downloadable personal health records, and electronic documents produced by health systems. The rates at which individuals are accessing their personal health information through patient portals are increasing substantially. Health Information National Trends Survey (HINTS), a nationally representative cross-sectional survey of 22 266 (2014, 2017–2020, 2022), showed that US adults' access to online medical record more than doubled over the past 8 years.² There was also a stark increase during COVID, moving from 46% in 2020 to 68% in 2022.

For children, 1 key piece of their personal health information is their immunization data. While state and federal health agencies have standard childhood immunization schedule forms that parents and caregivers can use to track their child's immunizations, parents and caregivers typically

collect and transcribe immunization dates from patient portals, downloaded personal health records, and electronic documents produced by health systems.^{3–5}

However, patient portals suffer from well-documented usability problems, including problems related to confusing terms and poor design of information displays.^{6–14} While usability testing of patient portals has not typically focused on interpretation of immunization information, several studies have assessed individuals' abilities to view lab results and track vitals, a parallel task involving longitudinal data. Two studies focused on use of portals by pediatric populations noted participant difficulty interpreting data and test results, and understanding medical terminology.^{15,16} Laboratory abbreviations, values, and results were particularly challenging for participants to interpret.¹⁶

Well-designed data visualizations can help patients understand and use their personal health information. Good data visualizations, if designed to leverage humans' quick and effective pre-attentive processing abilities (eg, differentiating colors, sizes, shapes, and spatial positions), can help

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individuals provide structure and organization to their perceptions, improve recall, and identify patterns.¹⁷ While patient-facing data visualizations have been shown to help individuals understand and interpret their health data,^{18,19} a recent systematic review found that data visualization research within healthcare has focused heavily on clinician or researcher-facing visualizations (834 clinician/researcher-facing studies vs 39-patient facing studies).¹⁸

Objective

In this study, we evaluated how well lay individuals could interpret and use 4 EHR patient-facing immunization

visualizations to complete an immunization form (Figure 1). The 4 visualizations contained the same de-identified immunization information for a 5-year old child and represented realistic ways individuals can currently access immunization information electronically: 2 EHR vendor patient portals (Epic MyChart and eClinicalworks), a downloadable copy of the patient's full record, and an electronic letter generated by a clinic using EHR data. Three of the 4 immunization visualizations shown in Figure 1 are in the form of tables (ECW, Lucy, and eLetter). Leading visualization designers see tables as a key type of visualization, but note that careful choices are often omitted from the table design process.²⁰ Recent publications within health informatics, including a systematic

Epic MyChart Patient Portal

DTaP ① Learn more Dates on file: 01/26/2018, 09/22/2016 View details	DTaP / Hep B / IPV ① Learn more Dates on file: 12/06/2016, 07/29/2016 View details	DTaP / IPV ① Learn more Dates on file: 06/05/2020 View details
Hep A, ped/adol ① Learn more Dates on file: 05/24/2018, 05/24/2017 View details	Hep B, ped/adol ① Learn more Dates on file: 05/20/2016 View details	Hib (PRP-T) ① Learn more Dates on file: 01/26/2018, 12/06/2016, 09/22/2016, 07/29/2016 View details
IPV ① Learn more Dates on file: 09/22/2016 View details	Influenza, injectable, quadrivalent, preservative free ① Learn more Dates on file: 10/12/2020, 09/29/2019, 11/24/2018, 11/20/2017 View details	Influenza, injectable, quadrivalent, preservative free, pediatric ① Learn more Dates on file: 01/11/2017, 12/06/2016 View details
MMR ① Learn more Dates on file: 05/24/2017 View details	MMRV ① Learn more Dates on file: 06/05/2020 View details	Pneumococcal Conjugate 13-Valent ① Learn more Dates on file: 08/08/2017, 12/06/2016, 09/22/2016, 07/29/2016 View details
Rotavirus Pentavalent ① Learn more Dates on file: 12/06/2016, 09/22/2016, 07/29/2016 View details	Varicella ① Learn more Dates on file: 08/08/2017 View details	

Epic Downloadable Lucy Summary

Immunizations

DTaP (Given 1/26/2018, 9/22/2016)
DTaP / Hep B / IPV (Given 12/6/2016, 7/29/2016)
DTaP / IPV (Given 6/5/2020)
Hep A, ped/adol (Given 5/24/2018, 5/24/2017)
Hep B, ped/adol (Given 5/20/2016)
Hib (PRP-T) (Given 1/26/2018, 12/6/2016, 9/22/2016, 7/29/2016)
IPV (Given 9/22/2016)
Influenza, injectable, quadrivalent, preservative free (Given 10/12/2020, 9/29/2019, 11/24/2018, 11/20/2017)
Influenza, injectable, quadrivalent, preservative free, pediatric (Given 1/11/2017, 12/6/2016)
MMR (Given 5/24/2017)
MMRV (Given 6/5/2020)
Pneumococcal Conjugate 13-Valent (Given 8/8/2017, 12/6/2016, 9/22/2016, 7/29/2016)
Rotavirus Pentavalent (Given 12/6/2016, 9/22/2016, 7/29/2016)
Varicella (Given 8/8/2017)

Clinic-generated Letter Based on Epic Data

Immunizations				
DTaP	7/29/2016	9/22/2016	12/6/2016	1/26/2018
	6/5/2020			
Hep A	5/24/2017	5/24/2018		
Hep B	5/20/2016	7/29/2016	12/6/2016	
Hib	7/29/2016	9/22/2016	12/6/2016	1/26/2018
Influenza	12/6/2016	1/11/2017	11/20/2017	11/24/2018
	9/29/2019	10/12/2020		
MMR	5/24/2017	6/5/2020		
Pneumococcal Conjugate	7/29/2016	9/22/2016	12/6/2016	8/8/2017
Polio	7/29/2016	9/22/2016	12/6/2016	6/5/2020
Rotavirus	7/29/2016	9/22/2016	12/6/2016	
Varicella	8/8/2017	6/5/2020		

eClinicalWorks Patient Portal

Vaccine	Date Given
DTaP, <7 yrs (Infanrix)	07/29/2016
DTaP, <7 yrs (Infanrix)	09/22/2016
DTaP, <7 yrs (Infanrix)	12/06/2016
DTaP, <7 yrs (Infanrix)	01/26/2018
DTaP, <7 yrs (Infanrix)	06/05/2020
Hemophilus B vaccine, (PedvaxHIB)	07/29/2016
Hemophilus B vaccine, (PedvaxHIB)	09/22/2016
Hemophilus B vaccine, (PedvaxHIB)	12/06/2016
Hemophilus B vaccine, (PedvaxHIB)	01/26/2018
Hepatitis A pediatric, 2-dose (Havrix)	05/24/2017
Hepatitis A pediatric, 2-dose (Havrix)	05/24/2018
Hepatitis B, vaccine pediatric 3-dose (Engerix-B)	05/20/2016
Hepatitis B, vaccine pediatric 3-dose (Engerix-B)	07/29/2016
Hepatitis B, vaccine pediatric 3-dose (Engerix-B)	12/06/2016
Influenza vaccine, historical	12/06/2016
Influenza vaccine, historical	10/25/2021
MMR vaccine, live	05/24/2017
MMR vaccine, live	06/05/2020
Pneumococcal, 13 valent (Prevnam 13)	07/29/2016
Pneumococcal, 13 valent (Prevnam 13)	09/22/2016
Pneumococcal, 13 valent (Prevnam 13)	12/06/2016
Pneumococcal, 13 valent (Prevnam 13)	08/08/2017
Polio vaccine, (IPOL)	07/29/2016
Polio vaccine, (IPOL)	09/22/2016
Polio vaccine, (IPOL)	12/06/2016
Polio vaccine, (IPOL)	06/05/2020
rotavirus, unspecified formulation	07/29/2016
rotavirus, unspecified formulation	09/22/2016
rotavirus, unspecified formulation	12/06/2016
Varicella, (Varivax)	08/08/2017
Varicella, (Varivax)	06/05/2020

Figure 1. Four immunization visualizations, each containing the same information. © 2023 Epic Systems Corporation; © eClinicalWorks.

review of patient-facing visualizations, largely omit tables as a key type of health data visualization.^{18,21,22} The MyChart representation, while not in table format, is the most easy-to-access immunization data provided by the EHR vendor with the highest overall market share (Epic Systems Corporation).

While 3 of the 4 visualizations in this study are tables, they adhere to different design principles, shown in Table 1. In addition, the MyChart and Lucy visualizations display vaccines by the content of the shot (eg, DTap/HepB/IPV), while ECW and the eLetter display vaccines by immunization type (eg, HepB) and display dates in chronological order. One key difference between ECW and the eLetter is the names given to the vaccines (eg, “DTap” in the eLetter vs “DTap, <7 years [Infanrix]” in ECW) and the listing of 1 date per row in ECW vs multiple dates per row in the eLetter.

Participants used 1 of the 4 visualization-types to complete an immunization form modeled after 3 existing public health immunizations forms (Centers for Disease Control, Minnesota Department of Health, and California Department of Public Health). The form included dates when a child received 7 immunization-types (HepB, DTap, Hib, PCV, IPV, MMR, Chickenpox, and HepA) each with multiple doses (Figure 2). Figure 2 shows the MN, CA, CDC immunization forms as exemplars, and the form we used to collect data in this study. We were limited to using an IRB-approved secure data collection tool, specifically Qualtrics or REDCap. The form used in this study is not meant to be a suggested design, but rather a secure online Qualtrics form that mimics the structure of the CDC and state forms. We had 3 design choices to make when we created the Qualtrics study form: (1) what vaccines to include some states require fewer vaccines on their form (eg, CA), (2) what order to list the vaccines in (MN and CDC list them by the age of first dose; CA does not), and (3) the names of the vaccines (full names and/or abbreviations.) Table 2 compares the vaccine names and doses across the forms. Our form used the 8 vaccines on the CDC and MN forms, rather than the 6 on the CA form. We listed the vaccines based on the age of the first dose. We

showed the full name of the immunization, with the abbreviation in (), to ensure our form had both name formats.

We evaluated whether the visualization-type affected 2 primary outcome measures, as defined by Isenberg:²³ (1) performance (measured by how long it took to complete the immunization form and how many dates they entered correctly) and (2) user experience (measured by perceptions of task complexity). We also assessed whether several other factors influenced task performance, including whether participants had previously used electronic immunization data, whether they had completed this type of task before, their subjective numeracy and health literacy, and several demographic variables.

Materials and methods

Participants

We recruited participants using Amazon Mechanical Turk (mTurk) and randomly assigned participants to see 1 of 4 visualizations. mTurk has been shown to be an effective method for recruitment in health-related and data visualization design research.^{24,25} We over-recruited because we knew that the mTurk data would need to be screened for duplicate IP addresses, Captcha failures, and answers to a free-text question that did not make sense (all known mechanisms for cleaning mTurk data). After screening, we had complete data for 69 participants. These data were spread across the Epic patient portal (MyChart) ($n=16$), eClinicalWorks (ECW) patient portal ($n=16$), downloadable EHR summary (Lucy) ($n=15$), and clinic-generated letter based on EHR data (eLetter) ($n=22$). In a systematic review of 581 papers focus on visualization evaluation methods from 10 years of IEEE Visualization conference publications,²³ of those that included potential end users (113 of 581 studies), a large majority (92 studies) included 20 or fewer participants. Participants were dispersed geographically throughout the United States, with an additional 4 located outside of the United States.

Procedure

Participants read a study information sheet within Qualtrics and chose whether to proceed. If they continued, participants were randomly shown 1 of the 4 immunization visualizations. Remotely administered studies are thought to suffer more from participant fatigue, causing participants to become tired, bored, or unmotivated. Within-subjects designs can exacerbate this problem. In this study, if participants were to complete all 4 visualizations (based on preliminary data and confirmed by our findings) their session would have lasted roughly an hour. Between the likely occurrence of fatigue, and possible carryover effects, we chose a between-subjects design. Other published patient-facing visualization studies have used a similar between-subjects approach, randomizing participants into visualization conditions.¹⁸

We gave participants the following instructions: “The image below is from a doctor. It shows dates when a 5-year-old child received their immunizations (vaccines). Use the immunization data in the image to enter the correct vaccine dates in each field.” Participants used data from the visualization to enter 26 dates into the immunization form. Qualtrics captured the time participants spent on this task-related study page, a proxy for the time it took them to read the immunization data and enter it in the form. When the participants were

Table 1. Adherence to key table design principles.²⁰

Principle	Description	Lucy	ECW	eLetter
Column and row separation	Use white space alone whenever possible, subtle rules if need, and avoid grids	X	X	X
Row categories	Separate/group rows into categories		X	X
Row labels	Do not repeat row label when there are repeating row categories	X		X
Row ordering	Use alphabetical order when there is no natural sequence	X	X	X
Column divisions	Arrange entries across separate columns			X
Column ordering	Arrange time-series subdivisions across columns			X
Alignment	Left-align text, any alignment for dates		X	X
Date formatting	Use the same number of digits across dates		X	
Font	Use the same legible font throughout the table	X	X	X

MN Dept of Health

Immunization Form

Name _____ Birthdate _____

Enter the dates for each vaccine your child has received to date. Specify the month, day, and year of each dose such as 05/01/2010.

Vaccine	Birth to 6 months	12-24 months	At Kindergarten	At 7th grade	At 12th grade
Hepatitis B					
Diphtheria, Tetanus, Pertussis (DTaP, DT, Td)					
Haemophilus influenzae type b (Hib)					
Pneumococcal (PCV)					
Polio					
Measles, Mumps, Rubella (MMR)					
Chickenpox (varicella)					
Hepatitis A					
Tetanus, Diphtheria, Pertussis (Tdap)					
Meningococcal (MCV4)					

Minnesota law requires children enrolled in child care, early childhood education, or school to be immunized against certain diseases, unless the child is medically or non-medically exempt.

Instructions for parent or guardian:

- Fill out the dates in chronological order even if your child received a vaccine outside of the age/grade category that the box is in. Depending on the age of your child, they may not have received all vaccines; some boxes will be blank.
 - If you have a copy of your child's immunization history, you can attach a copy of it instead of completing the front of this form.
 - Your doctor or clinic can provide a copy of your child's immunization history. If you are missing or need information about your child's immunization history, talk to your doctor or call the Minnesota Immunization Information Connection (MIIC) at 651-263-3880 or 800-657-3970.
- Sign or get the signatures needed for the back of this form.
 - Document medical and/or non-medical exemptions in section 3.
 - Verify history of chickenpox (varicella) disease in section 2.
 - Provide consent to share immunization information (optional) in section 3.

mn DEPARTMENT OF HEALTH
Immunization Program (2023)
www.health.state.mn.us/immunize

CA Dept of Public Health

CALIFORNIA PRE-KINDERGARTEN AND SCHOOL IMMUNIZATION RECORD

Pre-kindergarten facility and school staff must record the required vaccine dose information and status of requirements for each pupil. See reverse side for guidance.

PUPIL NAME (LAST, FIRST, MIDDLE) _____ DATE OF BIRTH (MM/DD/YYYY) _____

STUDENT ID _____ SEX _____

RACE _____

ETHNICITY _____

NAME OF PARENT/GUARDIAN (LAST, FIRST) _____ BIRTHDATE (MM/DD/YYYY) _____

REQUIRED VACCINE	DATE EACH DOSE WAS GIVEN (MM/DD/YYYY)	Permanent Medical Exemption	Notes for School Requirements			
	1 st	2 nd	3 rd	4 th	5 th	
IPV / OPV (polio)						4 doses meet TKK-12 requirement, as do 3 doses, if 2 nd dose given at age 24 years.
DTaP / DTP — Age 0-6 years						5 doses meet TKK-12 requirement, as do 4 doses, if 2 nd dose given at age 24 years.
Tdap / Td — Age 7+ years (Diphtheria, Tetanus, Pertussis)						3 doses, if 2 nd Tdap dose at age 24 years. Tdap dose may meet 7 th Grade requirement.
MMR (Measles, Mumps, Rubella)						2 doses meet TKK-12 requirement. Doses must be given at age 24 years.
Hib (Haemophilus influenzae type b)						Required for pre-kindergarten only. At least 1 dose must be given at age 24 years.
Hep B (Hepatitis B)						3 doses meet TKK-12 requirement.
VAR / VZV (Varicella or Chickenpox)						2 doses meet TKK-12 requirement.
Tdap — 7 th Grade (Tetanus, Diphtheria, Pertussis)						1 dose given at age 27 years meets requirement for 7 th grade advancement and 7 th -12 th grade advancement.

STATUS OF REQUIREMENTS	Staff Initials (if required, pupil's immunization record)	Has All Required Vaccine Doses	Temporary Medical Exemption	Requires Follow-up	Missing Doses Not Currently Due—Conditional	Missing Doses Are Overdue—Needs Dose Now	Follow-up (Date) (See conditional exemption schedule or exemption end date)	Other (See codes on reverse side)	Date Requirements Met
Pre-Kindergarten (child care or preschool)								SEP, PRE (pre-2016)	
TKK-12								SEP, PRE, TKO, HOME, PRE (pre-2016)	
7 th Grade (Advancement or admission)								SEP, PRE, TKO, HOME	

No California Department of Public Health documents are controlled by the gathering and use of personally identifiable data. Personal information is not disclosed, made available, or otherwise used for purposes other than those specified in the use of collection, except with consent or as authorized by law or regulation. The Department's information management practices are consistent with the Information Practices Act (Civil Code Section 1798 et seq.), the Public Records Act (Government Code Section 6250 et seq.), Government Code Sections 11101.5 and 11101.6, and with other applicable laws pertaining to information privacy.

CDPH 248 (1/19)

CDC

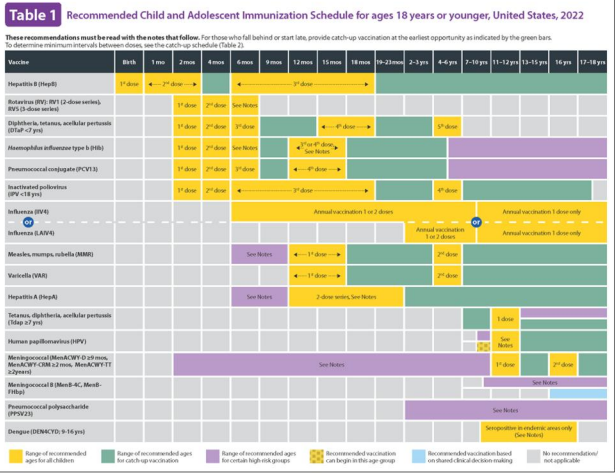


Figure 2. MN, CA, CDC, and our immunization forms.

satisfied with the dates they entered, they proceeded to the next screen to answer the remainder of the study questions.

Participants completed the validated 7-point Single Ease Question (SEQ) and the validated 6-question NASA-TLX

Our Form

First Dose Second Dose Third Dose

Hepatitis B (Hep B)

First Dose Second Dose Third Dose Fourth Dose Fifth Dose

Diphtheria, Tetanus, Pertussis (DTaP)

First Dose Second Dose Third Dose Fourth Dose

Haemophilus influenzae type b (Hib)

First Dose Second Dose Third Dose Fourth Dose

Pneumococcal (PCV)

First Dose Second Dose Third Dose Fourth Dose

Polio (IPV)

First Dose Second Dose

Measles, Mumps, Rubella (MMR)

First Dose Second Dose

Chickenpox (Varicella)

First Dose Second Dose

Hepatitis A (Hep A)

instrument measuring perceived workload.^{26–29} The SEQ assesses how difficult users find a task, based on the single question “Overall, how difficult or easy was the task to complete?” NASA-TLX measures perceived workload across 6

Table 2. Immunization naming conventions across forms.

Immunization name used in our study form	Number of doses	MN Department of Health form immunization name	California Department of Public Health form immunization name	CDC form immunization name
Hepatitis B (Hep B)	3	Hepatitis B	Hep B (Hepatitis B)	Hepatitis B (HepB)
Diphtheria, Tetanus, Pertussis (DTaP)	5	Diphtheria, tetanus, pertussis (DTaP, DT, Td)	DTaP/DTP—Age 0–6 years Tdap/Td—Age 7+ years (diphtheria, tetanus, pertussis)	Diphtheria, tetanus, acellular pertussis (DTaP <7 years)
Hemophilus influenzae type b (Hib)	4	Hemophilus influenzae type b (Hib)	Hib (Hemophilus influenzae type b)	Hemophilus influenzae type b (Hib)
Pneumococcal (PCV)	4	Pneumococcal (PCV)	<i>Not on form</i>	Pneumococcal conjugate (PCV13)
Polio (IPV)	4	Polio	IPV/OPV (Polio)	Inactivated poliovirus (IPV <18 years)
Measles, mumps, rubella (MMR)	2	Measles, mumps, rubella (MMR)	Measles, mumps, rubella (MMR)	Measles, mumps, rubella (MMR)
Chickenpox (varicella)	2	Chickenpox (varicella)	VAR/VZV (varicella or chickenpox)	Varicella (VAR)
Hepatitis A (Hep A)	2	Hepatitis A (Hep A)	<i>Not on form</i>	Hepatitis A (HepA)

domains: mental demand, physical demand, temporal demand, performance, effort, and frustration. The 6 questions can be aggregated into a single workload measure. To assess their familiarity with the task, we asked participants whether they had ever viewed their health data electronically, viewed immunization data electronically, and/or filled out an immunization form similar to the one in this study.

We measured numeracy using the 8-item Subjective Numeracy Scale (SNS).^{30,31} The SNS is a self-report measure of numerical ability and preference (4 questions each) containing no mathematical questions and has no correct or incorrect answers, each question answered on a 6-point scale, with 6 corresponding to the highest level of numeracy. The SNS has been validated against objective numeracy measures. We assessed health literacy using Chew's 3 health literacy screening questions, "How often do you have someone help you read hospital materials?" "How confident are you filling out medical forms by yourself?" and "How often do you have problems learning about your medical condition because of difficulty understanding written information?"³² These brief questions have been shown to be effective in detecting inadequate health literacy. Numeracy and health literacy are known to influence individuals' abilities to interpret visualizations, but they are often not measured in visualization evaluation studies.^{18,21,22} We then asked participants several demographic questions and asked for open-ended feedback on how to improve the visualizations.

Outcome measures and statistical analysis

Our primary outcome measures were the time taken to complete the task, the number of correct dates entered by the participant, and the perceived task difficulty. Our main research questions were:

- 1) Does immunization visualization design affect the time participants take to complete the form?
- 2) Does immunization visualization design affect the number of dates participants correctly enter on the form?
 - a) Requiring the dates be in chronological order, as was asked in the task
 - b) Allowing the dates to be in any order (eg, reverse chronological order)

- 3) Does immunization visualization design affect perceived task difficulty?

We also assessed whether numeracy, health literacy, demographic factors, and familiarity with the study task affected these outcome measures. All outcome measures had non-normal distributions, so statistical tests were completed using the independent-samples Kruskal-Wallis test, Mann-Whitney *U* test for 2 samples, and Spearman's rank correlations. Statistical analyses were performed using IBM SPSS Statistics (Version 29).

Ethics

This study was approved by the University of Minnesota IRB.

Results

Impact of immunization visualization design on task performance

Table 3 summarizes the impact of visualization design on our performance measures, including task time and number of correct dates out of 26 possible.

Does the visualization design affect the time participants take to enter the dates on the form?

A Kruskal-Wallis test showed that there was a difference in the time taken to complete the task based on the visualization type, $\chi^2(3) = 14.928$, $P = .002$. Pairwise comparison of groups and adjustment for the Bonferroni correction showed that those using MyChart took significantly longer than those using the eLetter ($P = .008$) or ECW ($P = .029$).

Does the visualization design affect the number of dates the participants correctly enter the dates on the form?

When requiring that dates be both correct and in chronological order, as was asked in the task, Kruskal-Wallis tests indicate that there is a difference in the number of correct dates based on the visualization type, $\chi^2(3) = 20.252$, $P < .001$. Pairwise comparison of groups and adjustment for the Bonferroni correction showed that those using MyChart entered fewer correct dates than those using the eLetter ($P < .001$) or ECW ($P = .018$) and those using the Lucy summary entered fewer correct dates than those using the eLetter ($P = .017$).

Table 3. Performance outcomes by visualization-type.

Visualization Design	<i>n</i>	Task time		Avg. number of correct dates in chronological order		Avg. number of correct dates in any date order	
		Mean	<i>P</i> -value	Mean	<i>P</i> -value	Mean	<i>P</i> -value
ECW	16	11.05	.002	25.19	.001	25.44	.015
eLetter	22	11.47		25.27		25.36	
Lucy	15	16.82		22.93		25.07	
MyChart	16	26.30		19.75		23.00	
Significant Pairwise Comparisons		MyChart>eLetter	.008	MyChart < eLetter	.001	MyChart < eLetter	.015
		MyChart>ECW	.029	MyChart < ECW	.018		
				Lucy < eLetter	.017		

Table 4. Correlation between subjective numeracy and task performance.

	Task time		Number of correct dates in chronological order		Number of correct dates regardless of the date order	
	Correlation coefficient	<i>P</i> -value	Correlation coefficient	<i>P</i> -value	Correlation coefficient	<i>P</i> -value
Subjective numeracy ability	−0.042	.731	0.025	.836	−0.057	.644
Subjective numeracy preference	−0.104	.394	0.156	.200	0.114	.353

When requiring that dates be correct but in any date order (eg, reverse chronological order), Kruskal-Wallis tests indicate that there was still a difference in the number of correct dates based on the visualization type, $\chi^2(3) = 10.477$, $P < .015$. Pairwise comparison of groups and adjustment for the Bonferroni correction showed that those using MyChart entered fewer correct dates than those using the eLetter ($P = .015$).

Spearman's rank correlation showed a significant negative correlation between the time taken to complete the task and the number of dates entered correctly when required to be in chronological order, $r(67) = -0.347$, $P = .004$, and when allowed to be in any date order, $r(67) = -0.357$, $P = .003$.

Does the visualization design affect the perceived task difficulty, based on the SEQ and NASA-TLX scales?

Workload measures were consistent across domains, with SEQ statistically significantly correlated to all NASA-TLX measures and all NASA-TLX individual domains statistically significantly correlated to each other. Kruskal-Wallis tests showed that there was no difference in the perceived task difficulty based on the visualization design.

Impact of numeracy, health literacy, and demographics on task performance

The SNS is comprised of an ability subscale and preference subscale. For our participants, the SNS ability subscale and SNS preference subscale were positively correlated, $r(67) = 0.404$, $P < .001$. Average SNS ability score was 3.71 out of 6 and the average SNS preference score was 3.87 (6 being the highest level of numeracy). SNS scores were not significantly different across the 4 groups ($P = .464$ for ability and $P = .377$ for preference). Spearman's rank correlation analysis showed that neither the SNS ability subscale or SNS preference subscale were correlated with any of the 3 task performance measures, shown in Table 4.

Kruskal-Wallis tests showed that overall health literacy scores were not significantly different across the 4 groups ($P = .47$)

and only one relationship between the health literacy questions and performance measures was significant, as shown in Table 4, but there was not any directionality in the response pattern. Kruskal-Wallis tests analyzing the relationship between demographic variables and measures of experiences with the task, and performance measures, summarized in Table 5, show that no relationships were significant except the number of children a participant had (worse performance with more children). We include categories in the table with at least 1 participant.

Participant qualitative feedback

Group by immunization-type, not shot composition

MyChart and Lucy displayed and grouped immunizations by shot composition (eg, DTaP/HepB/IPV). This required users to look across multiple boxes or rows to find all doses of 1 immunization type. For example, within MyChart and Lucy, the user needed to look across 3 boxes or rows to find all DTaP dates. Individuals using MyChart and Lucy also need to decipher that MMRV is a shot containing both MMR and Varicella. The eLetter and ECW separated and grouped immunizations by immunization-type. Several participants commented about restructuring these groupings. For example:

Just have each vaccine written separately with the dates. (MyChart)

I found it frustrating that the bunched combo vaccines were listed together and not separately. (Lucy)

List all the dates for the particular vaccination together. . . Example: chicken pox was listed under Varicella and MMRV—break out dates under MMR and varicella separately. (Lucy)

Standardize and simplify labels

The eLetter sometimes used common names (eg, Polio alone instead of IPV) and sometimes used abbreviation/clinical

Table 5. Impact of numeracy, literacy, and demographic variables on task performance.

	<i>n</i>	Task time		Avg. number of correct dates in chronological order		Avg. number of correct dates regardless of the date order	
		Mean	<i>P</i> -value	Mean	<i>P</i> -value	Mean	<i>P</i> -value
Health literacy							
How often do you have problems learning about your medical condition because of difficulty understanding written information?			.384		.376		.180
Always	3	11.67		25.67		25.67	
Often	3	14.20		20.00		26.00	
Sometimes	17	16.10		24.71		25.05	
Occasionally	25	21.00		21.56		23.72	
Never	21	10.77		24.90		25.48	
How often do you have someone (like a family member, friend, or hospital worker) help you read hospital materials?			.122		.139		.024*
Always	2	16.42		26.00		26.00	
Often	10	14.03		24.80		25.20	
Sometimes	13	24.17		20.15		22.23	
Occasionally	17	18.03		23.06		25.47	
Never	27	11.43		24.63		25.30	
How confident are you at filling out medical forms by yourself?			.251		.676		.238
Extremely	27	11.01		24.30		25.41	
Quite a bit	18	15.17		23.11		24.50	
Somewhat	19	24.52		23.11		23.84	
Not at all	5	13.22		21.60		25.80	
Demographic variables							
Age			.225		.351		.247
18-24	1	6.26		26.00		26.00	
25-44	52	13.62		23.71		24.88	
45-64	15	24.81		22.27		24.20	
65-84	1	15.93		26.00		26.00	
Gender			.167		.327		.356
Female	35	18.10		23.51		24.23	
Male	32	14.29		23.25		25.28	
Non-binary	2	5.81		26.00		26.00	
Race			.840		.273		.129
Asian	4	13.38		24.25		25.75	
Black/African American	5	16.81		25.40		25.80	
Multiple races	3	14.20		25.33		25.33	
Native American/Alaskan	3	14.70		26.00		26.00	
White	54	16.26		22.98		24.50	
Ethnicity			.868		.516		.470
Hispanic/Latinx	6	11.35		22.17		25.50	
Non-Hispanic/Latinx	62	16.44		23.55		24.68	
Prefer not to say	1	15.17		26.00		26.00	
Education			.176		.197		.092
High school	9	11.42		25.00		25.78	
Associates degree	4	8.03		26.00		26.00	
Technical degrees	2	17.97		24.50		24.50	
Bachelor's degree	30	12.84		23.70		25.00	
Master's degree	23	23.25		21.96		23.87	
Doctoral degree	1	11.72		25.00		25.00	
Employment status			.689		.435		.423
Employed	62	16.61		23.21		24.66	
Unemployed	5	10.13		25.60		25.60	
Retired	1	15.93		26.00		26.00	
Prefer not to say	1	6.26		26.00		26.00	
Experience as a healthcare worker*			.958		.558		.932
Yes	21	15.27		23.76		24.14	
No	48	16.29		23.33		25.04	

(continued)

Table 5. (continued)

	<i>n</i>	Task time		Avg. number of correct dates in chronological order		Avg. number of correct dates regardless of the date order	
		Mean	<i>P</i> -value	Mean	<i>P</i> -value	Mean	<i>P</i> -value
Income			.847		.780		.264
Less than \$25 000	7	11.40		24.57		25.29	
\$25 000-49 999	13	17.72		23.38		25.08	
\$50 000-74 999	20	17.48		23.60		25.50	
\$75 000-99 999	17	15.93		23.35		23.71	
\$100 000-149 999	4	21.35		21.00		22.50	
\$150 000 or more	7	10.68		23.57		25.43	
Prefer not to say	1	11.72		25.00		25.00	
Marital status			.562		.111		.208
Single	18	11.65		25.56		25.67	
Married	47	17.61		22.64		24.40	
Partnered	1	19.87		25.00		25.00	
Divorced	3	15.07		23.33		25.00	
Number of children			.025*		.014*		.266
0	25	10.52		25.28		25.40	
1	21	19.11		23.19		24.38	
2	19	18.22		22.42		24.79	
3	4	23.01		18.50		22.75	
Viewed health data electronically*			.663		.292		.313
Yes	49	16.92		22.94		24.55	
No	20	13.67		24.75		25.30	
Viewed immunization data electronically (if yes to above)*			.565		.558		.566
Yes	33	14.57		22.55		24.45	
No	16	21.77		23.75		25.30	
Filled out a form like the one in this study before			.839		.192		.570
Yes	26	15.90		21.08		23.96	
No	38	16.56		24.84		25.24	
I'm not sure	5	11.95		25.40		25.40	

All analyses use the Kruskal-Wallis ANOVA test, except those noted as (*), which use the Mann-Whitney *U* test for 2 samples.

names (eg, DTap instead of diphtheria, tetanus, pertussis). Some visualizations included extraneous information. ECW sometimes included manufacturer or dose amount, such as “Hepatitis B, vaccine pediatric 3-dose (Engerix-B)” and Lucy and MyChart used “HepA, ped/adol.” Two participants commented on these naming conventions:

some of the names used abbreviations or vice-versa. it would be nice if they were all consistent. (MyChart)

It would be a little easier if the names were always identical for the data entry spot. (ECW)

List dates in chronological order from oldest to most recent

Childhood immunization schedules are based on age ranges, from birth onward. MyChart and Lucy list dates in reverse chronological order, starting with the most recent immunization, which caused confusion for some participants. Several participants commented about the ordering of dates:

Organize the dates in order. (MyChart)

It was annoying that the vaccines were listed from most recent to least recent. (Lucy)

It would be ideal and easy to have the dates arranged and displayed in chronological order. (Lucy)

Leverage good table design principles

The MyChart grid format required significant visual scanning to identify immunization names and dates. In general, grid views are good for presentation of images and supporting visual comprehension and differentiation.³³ Tables are generally better for presentation of data and supporting numeric or reading comprehension.^{20,34} Several participants provided possible improvements related to table layout:

columns that are distinguished separately from each other would be more visually organized. (Lucy)

Grids/tables are always helpful. (Letter)

If the dates were put rows with lines to separate them, it would be easier to see the right dates. (Letter)

Order the immunization list thoughtfully

Two participants commented that they would prefer the vaccines be listed in alphabetical order:

Alphabetical might make it easier to correspond data.
(ECW)

It would be ideal and easy to have ... the vaccines arranged in alphabetical order. (Lucy)

Discussion

Study participants used 4 immunization visualization designs to complete a childhood immunization form. It took on average 15.98 min to complete the form. Of 26 total dates, participants entered, on average, 23.46 correct dates in chronological order. Allowing for any date order, participants entered, on average, 24.18 correct dates.

The immunization visualization design, however, significantly impacted all performance measures. Those using MyChart took statistically significantly longer to complete the task (mean = 26.30 min) than those using the eLetter (mean = 11.47 min) or ECW (mean = 11.05 min). When requiring dates to be both correct and in chronological order, those using MyChart entered statistically significantly fewer correct dates (mean = 19.75 correct) than those using the eLetter (mean = 25.27 correct) or ECW (mean = 25.19 correct). When requiring dates to be correct, but allowing any order, those using MyChart still entered statistically significantly fewer correct dates (mean = 23.00 correct) than those using the eLetter (mean = 25.36 correct).

While the task in this study might seem straightforward, successfully completing this task required 4 of the lower-level outcomes described by Ancker et al.²²: (1) identification of relevant data (ie, finding dates), 2) discrimination (ie, identifying differences in dates), 3) categorization of data (ie, categorizing dates by immunization-type), and 4) contrast (ie, ability to order dates in chronological order). The 4 visualization designs had unique design elements that may have contributed to these performance differences, via their support (or lack of support) of these lower-level outcomes. If it is not possible to auto-generate a customized form and display it to patients, we suggest several design principles that patient-facing immunization visualizations should abide by, all of which are supported by known design guidelines and comments from participants.

- a) Placing dates for immunization-types together conveys that those groupings are meaningfully related. Placing meaningfully related elements in a visual display close together aligns with the Gestalt principle of proximity.³⁵ MyChart and Lucy did not adhere to this principle.
- b) Labels should use consistent, simple naming conventions. Immunization visualizations could use the format “common name (abbreviation/clinical name),” with no additional text. These recommendation align with Nielsen’s web design guidelines, specifically noting that designers should “Spell out abbreviations, initialisms, and acronyms, and immediately follow them by the abbreviation, in the first instance,” “Use customer-focused language” and “Use standard abbreviations”.³⁶ ECW, MyChart, and Lucy were not consistent in doing so.
- c) Childhood immunization schedules are based on age ranges, from birth onward. Similarly, immunization visualization dates should be listed from birth onward. For

a small set of dates, such as the case here, forward chronological ordering tells a cohesive data “story”. MyChart and Lucy list dates in reverse chronological order.

- d) Immunizations should be visualized in table format, similar to the immunization forms. Example table design principles include:
 - i) Increase the data-to-ink ratio by minimizing non-data elements (eg, extraneous lines and colors).^{34,37}
 - ii) Each row should contain 1 immunization type and multiple dates, as time-series categories should be across columns.^{34,38}
 - iii) Left-align text in each column, as it is easier to read than centered or right-aligned text.^{34,38}
 - iv) Provide enough spacing between columns and rows so the reader can easily distinguish between immunization types and dates.³⁴
- e) Order the immunization types in alphabetical order

The design of Table visualizations, that is visualizations of datasets expressed as Tables,³⁹ is crucial to effectively communicate quantitative information,²⁰ yet recent publications within health informatics, including a systematic review of patient-facing visualizations, largely omit discussion of tables.^{18,21,22} Future research should pay key attention to the value of tables for conveying health information, and conduct studies to understand the design principles that increase their effectiveness. Ideally, these studies would result in table style guides, templates, and repositories.²¹

Study limitations

We did not purposively sample participants based on demographic characteristics. Given the significant differences in performance across the 4 visualizations, future research should examine whether these differences are consistent across individuals from a more diverse sample. While we did have dispersed levels of health literacy and modest levels of numeracy, we did not target participants with low health literacy or numeracy, and suggest doing so in the future as these factors are known to influence individuals’ abilities to interpret visualizations.^{18,21,22}

The remote nature of our study caused some limitations. We could not view what participants were doing as they completed the task and we could not conduct debriefing interviews. Given that remotely administered studies are thought to suffer more from participant fatigue and the relatively long time taken to complete each task, we focused on the situation where the individual was up-to-date on their schedule. Future work should identify whether these differences exist in a situation where an individual is not up-to-date on their vaccinations.

While other patient-facing visualization studies have used a similar between-subjects approach to reduce participant fatigue and learning effects, this approach did reduce our sample size of individuals who viewed each visualization.¹⁸ If each participant interacted with all visualizations, our sample size may have been large enough to better detect differences across demographic characteristics, and health literacy and numeracy levels.

We limited our study to laptop or desktop viewing of these visualizations. Future research will replicate this work using mobile device versions of these visualizations.

Conclusion

This study demonstrated that the choice of immunization visualization design significantly affects individuals' performance completing a common task, measured by task time and accuracy. There was no difference in perceived workload across visualization designs. Based on our findings, we provided several practical guidelines for the design of immunization visualizations. Future research should focus on understanding the contexts of use and design elements that make tables an effective type of health data visualization.

Author contributions

All authors contributed to the design of the study, analysis of the results, drafting and revising the article, and the final approval of the submitted version.

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Conflicts of interest

None declared.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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